Prodigy Anesthesia Quick Review

Antiepileptic Drugs

What are the drugs of choice used in the treatment of status epilepticus?	Phenobarbital, phenytoin, benzodiazepines, thiopental, and propofol are appropriate choices for initial treatment. *Barash PG, Cullen BF, Stoelting RK, Cahalan MK, Stock MC, Ortega R. Clinical Anesthesia. 7th ed. Philadelphia, PA: Lippincott Williams and Wilkins; 2013: 620.*
Antiepileptic drugs can cause what adverse hematologic reactions? Which drugs are usually associated with these reactions?	Antiepileptic drugs are associated with anemias. The symptoms are typically mild, but can occasionally produce aplastic anemia. They are most commonly seen with valproate, carbamazepine, and phenytoin. *Hines RL, Marschall KE. Anesthesia & Co-Existing Diseases. 6th ed. Philadelphia, PA: Elsevier-Saunders; 2012: 251.*
What is the clinical use of phenytoin?	It is used to treat generalized tonic-clonic seizures, as well as partial seizures. *Katzung BG, Masters, SB, & Trevor AJ. Basic and Clinical Pharmacology. 12th ed. New York: McGraw-Hill; 2012: 408.*
What physiologic systems are affected by phenytoin?	Phenytoin alters membrane potentials and the concentrations of the neurotransmitters GABA, norepinephrine, and acetylcholine. It also interferes with the conductance of sodium, potassium, and calcium. *Katzung BG, Masters, SB, & Trevor AJ. Basic and Clinical Pharmacology. 12th ed. New York: McGraw-Hill; 2012: 403.*
List side effects associated with the use of phenytoin.	Gingival hyperplasia, hirsutism, nystagmus, diplopia, ataxia, hypotension, aplastic anemia, cardiac arrhythmias, Stevens-Johnson syndrome, and erythema multiforme. *Hines RL, Marschall KE. Anesthesia & Co-Existing Diseases. 6th ed. Philadelphia, PA: Elsevier-Saunders; 2012: 250.* *Katzung BG, Masters, SB, & Trevor AJ. Basic and Clinical Pharmacology. 12th ed. New York: McGraw-Hill; 2012: 409.*
What is phenytoin's main mechanism of action at therapeutic concentrations?	Phenytoin exerts its effects through the blocking of sodium channels and the inhibition of rapidly firing action potentials. *Katzung BG, Masters, SB, & Trevor AJ. Basic and Clinical Pharmacology. 12th ed. New York: McGraw-Hill; 2012: 408.*
What is the therapeutic blood level of phenytoin?	10-20 mcg/ml *Katzung BG, Masters, SB, & Trevor AJ. Basic and Clinical Pharmacology. 12th ed. New York: McGraw-Hill; 2012: 408.*
Describe the result of an intra-arterial injection or extravasation of phenytoin.	Marked vasoconstriction occurs, which can progress to compartment syndrome, gangrene, and skin necrosis. *Hines RL, Marschall KE. Anesthesia & Co-Existing Diseases. 6th ed. Philadelphia, PA: Elsevier-Saunders; 2012: 250.*

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What are the clinical uses of carbamazepine?	The original purpose for carbamazepine was in the treatment of trigeminal neuralgia, however, it is also used for the treatment of epilepsy and bipolar disorder. *Katzung BG, Masters, SB, & Trevor AJ. Basic and Clinical Pharmacology. 12th ed. New York: McGraw-Hill; 2012: 410.*
What is carbamazepine's mechanism of action?	Carbamazepine (Tegretol) blocks sodium channels and inhibits high-frequency neuronal firing. It also decreases synaptic transmission. *Katzung BG, Masters, SB, & Trevor AJ. Basic and Clinical Pharmacology. 12th ed. New York: McGraw-Hill; 2012: 409-410.*
How is carbamazepine administered?	Carbamazepine can only be administered orally. *Katzung BG, Masters, SB, & Trevor AJ. Basic and Clinical Pharmacology. 11th ed. New York: McGraw-Hill; 2009: 406.*
What are the most common adverse effects associated with carbamazepine?	Ataxia and diplopia, both of which are considered to be dose-related. *Katzung BG, Masters, SB, & Trevor AJ. Basic and Clinical Pharmacology. 12th ed. New York: McGraw-Hill; 2012: 410.*
What is the mechanism of action of Lamotrigine?	Lamotrigine exerts its effects through the depression of rapidly firing neurons, it decreases the release of glutamate at synapses, and it also inactivates sodium channels. *Katzung BG, Masters, SB, & Trevor AJ. Basic and Clinical Pharmacology. 12th ed. New York: McGraw-Hill; 2012: 414.*
List the adverse effects of Lamotrigine.	Nausea, diplopia, dizziness, headache, skin rash, and somnolence. *Katzung BG, Masters, SB, & Trevor AJ. Basic and Clinical Pharmacology. 12th ed. New York: McGraw-Hill; 2012: 414.*
How does gabapentin exert its antiepileptic effects?	It decreases the release of glutamate from synapses. *Katzung BG, Masters, SB, & Trevor AJ. Basic and Clinical Pharmacology. 12th ed. New York: McGraw-Hill; 2012: 413.*
What are the most common adverse effects seen with the use of gabapentin?	Dizziness, tremor, headache, ataxia, and somnolence *Katzung BG, Masters, SB, & Trevor AJ. Basic and Clinical Pharmacology. 12th ed. New York: McGraw-Hill; 2012: 413.*
Does gabapentin cause enzyme induction?	No. Unlike other anticonvulsants, gabapentin does not cause enzyme induction. *Katzung BG, Masters, SB, & Trevor AJ. Basic and Clinical Pharmacology. 12th ed. New York: McGraw-Hill; 2012: 413.*

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What is the most significant adverse effect of the antiseizure medication valproate?	It can produce hepatotoxicity. The risk is highest in patients under 2 years of age and patients taking multiple medications. Liver enzymes should be monitored closely in patients taking valproate. *Katzung BG, Masters, SB, & Trevor AJ. Basic and Clinical Pharmacology. 12th ed. New York: McGraw-Hill; 2012: 419.*
What is the therapeutic blood level of phenobarbital?	10 mcg/ml to 40 mcg/ml *Katzung BG, Masters, SB, & Trevor AJ. Basic and Clinical Pharmacology. 12th ed. New York: McGraw-Hill; 2012: 411.*

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